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DATAMINING Project

Third Review

Student Engagement Analysis and Achievement Prediction

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Presentation Outline

1. Project Overview & Abstract
2. Dataset Description
3. Data Preprocessing (WEKA) Preprocessing
4. Classification – J48 Decision Tree
5. Clustering – K-Means Algorithm
6. Results & Insights
7. Conclusion

PROJECT OVERVIEW

PROBLEM STATEMENT

Traditional student performance analysis is manual, time-consuming, and error-prone. It fails to identify hidden patterns in student engagement data. Hence, an automated system is needed to analyze data and predict student achievement effectively.

OBJECTIVES

- To analyze student engagement data to understand performance patterns
- To preprocess and transform the dataset using WEKA
- To build a classification model using the J48 decision tree algorithm
- To predict student achievement based on engagement factors

DATA SET DESCRIPTION

DATASET DESCRIPTION

• The dataset contains **student engagement data** collected for academic analysis
• It includes **100 instances and multiple attributes** related to activities
• Engagement is measured based on **participation, motivation, and time investment**
• A total of **key features are used to predict student achievement**

Dataset Characteristics

• Dataset name: **student_engagement_data.csv**
• Source: **Academic dataset (for analysis purpose)**
• Format: **CSV (Comma Separated Values)**
• Target attribute: **achievement**
• Contains both **numeric and nominal attributes**

Key Attributes

• **participation** and **category of activity**
• **motivation** and **prior experience**
• **time investment** and **frequency of participation**
• **role** and **mentorship support**
• **achievement** *(target variable)*

DATA PREPROCESSING PIPELINE

- Import **student_engagement_data.csv** into WEKA Explorer
- Examine dataset and identify attribute types (numeric / nominal)
- Remove unnecessary attributes such as **StudentID**
- Handle missing values using **ReplaceMissingValues** filter
- Apply mean for numeric and mode for nominal attributes
- Convert categorical data into binary using **NominalToBinary**
- Normalize numeric values to range **[0, 1]**
- Set **achievement** as the target class attribute
- Prepare cleaned dataset for classification using J48 algorithm

CLASSIFICATION – J48 DECISION TREE

About J48 Algorithm

- Java implementation of **C4.5 decision tree algorithm**
- Builds tree using **student engagement attributes as nodes**
- Works with both **numeric and categorical data**
- Provides **easy-to-understand and interpretable results**

Implementation Steps

- Open **Classify tab** in WEKA Explorer
- Select **J48 algorithm under Trees**
- Set **10-fold cross-validation** for evaluation
- Run the model to generate **decision tree for achievement prediction**

Key Results

- Model predicts **student achievement levels**
- Important attributes: **participation, motivation, role**
- Generates **clear decision rules for classification**
- Helps in deriving **insights on student performance**

10-FOLD CROSS VALIDATION

How It Works

- The student engagement dataset is divided into **10 equal subsets (folds)**
- Each subset is used as a **test set one time**
- The remaining subsets are used for **training the J48 model**
- This process is repeated **10 times for complete evaluation**

Unbiased Evaluation

- Each student record is **evaluated exactly once**
- Ensures **fair and unbiased performance measurement**
- Prevents duplication of test data

Overfitting Prevention

- Model is tested on **unseen engagement data in each iteration**
- Helps in improving **generalization of the model**

Reliable Accuracy

- Final accuracy is calculated as **average of all 10 iterations**
- Provides **consistent and stable results**

Efficient Use of Data

- All **100 student records** are utilized
- Data is used for both **training and testing efficiently**

CLUSTERING — K-MEANS ALGORITHM

K-Means is an unsupervised learning algorithm used to group **student engagement data** into clusters based on similarity in participation, motivation, and activity patterns.

Cluster Results

Cluster 0 – High Engagement

Students with **high participation, motivation, and time investment**

Cluster 1 – Moderate Engagement

Students with **average involvement and moderate activity levels**

Cluster 2 – Low Engagement

Students with **low participation and minimal involvement**

Process Overview

01 – Clustering Setup

Opened **Cluster tab in WEKA** and selected **SimpleKMeans algorithm**

02 – Set K = 3

Defined number of clusters as **3 to represent engagement levels**

03 – Run Algorithm

Performed clustering using **Euclidean distance to calculate centroids**

04 – Visualization

Observed grouping of students based on **engagement patterns across clusters**

RESULTS & KEY INSIGHTS

CLASSIFICATION INSIGHTS

Key Factor

• **Participation level** is the main factor affecting student achievement

Supporting Factors

• **Motivation, role, and time investment** also influence performance

Model Performance

• J48 model provides **moderate accuracy with reliable classification results**

Decision Rules

• Clear rules generated to **predict student achievement levels**

CLUSTERING INSIGHTS

High Engagement

• Students show **active participation and high involvement in activities**

Low Engagement

• Students have **low participation and minimal activity involvement**

Moderate Engagement

• Students show **average participation with mixed performance patterns**

Key Observation

• Clusters clearly differentiate **levels of student engagement and performance**

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CONCLUSION

- Student engagement dataset was successfully **preprocessed using WEKA filters** (Remove, ReplaceMissingValues, NominalToBinary, Normalize)
- J48 Decision Tree classifier was applied to **predict student achievement based on engagement factors**
- Important factors such as **participation, motivation, role, and time investment** significantly influence performance
- K-Means clustering grouped students into **high, moderate, and low engagement levels**
- The model achieved **27% accuracy**, indicating scope for improvement with better data and advanced algorithms
- Data-driven insights help educators **analyze student behavior and improve academic outcomes**
- The system can be extended for **real-time analysis and advanced predictive models**

THANK YOU